Answer Partitions and Oracle Access Determine Quantum Query Complexity

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An answer partition specifies which oracle instances share an output, while query complexity also depends on how those instances are accessed. We formulate exact one-query answer-partition resolution as block discrimination of a unitary oracle family. Applied to Deutsch's problem with a controlled response unitary V, the criterion shows that one query suffices if and only if -1 is an eigenvalue of V; cyclic addition in odd response dimension therefore raises the exact quantum cost to two queries, matching the classical cost while preserving the same computational-basis point values. Holding the standard Boolean point oracle fixed, we further show that balanced answer partitions with equal-sized cells can have different exact classical and quantum query complexities. A complete exact classification is given for all 35 balanced three-bit partitions, while the unresolved four-bit cases are reported explicitly as numerical candidates. These results separate the combinatorial specification of an answer from the oracle-dependent distinguishability that determines query complexity.

An oracle problem pairs an answer partition with an access model for the hidden instance. The partition groups instances into cells whose members share the same required answer. The access model determines what information each query makes available, classically or quantum mechanically, and how that information can be processed. Quantum query speedup therefore belongs to the task–oracle pair. We make this viewpoint precise and then test it on standard point oracles, response-unitary variants, and small exhaustive families of balanced tasks.

This viewpoint also sharpens the usual intuition about quantum parallelism. A quantum query prepares one coherent state, and the algorithm succeeds when interference makes the states associated with different answer cells distinguishable. The useful resource is consequently the structure of the oracle response—its phases, spectrum, accessible registers, and possible environment—rather than simply the formal number of branches in a superposition. Rewriting an oracle by fixed basis changes preserves this discrimination, whereas changing its response representation, aggregating information, or discarding a witness can alter it [1]. The examples that follow develop these ideas through exact one-query discrimination, parity, Bernstein–Vazirani, and a finite census of balanced partitions.

Let Ω be a finite instance set and let $g : \Omega \to J$ specify the required answer. The task induces the *answer partition*

$$\Pi_g = \{C_j : j \in g(\Omega)\}, \qquad C_j = g^{-1}(j). \tag{1}$$

This partition is indispensable bookkeeping, but it is not yet an access model or a complexity statement. In the query model the hidden instance labels a black-box transformation; the algorithm is not handed a state $|a\rangle$ on which the answer partition is a pre-existing observable. The operational question is whether the specified oracle and a fixed protocol can transfer the cell label to an accessible output.

More explicitly, let $\rho_a^{(T)}$ be the accessible state produced from instance a after T queries. Exact resolution means that there is one instance-independent measurement $\{P_j\}$ satisfying

$$P_j P_\ell = \delta_{j\ell} P_j, \; \sum_j P_j = I, \; \mathrm{Tr}(P_{g(a)} \rho_a^{(T)}) = 1. \tag{2}$$

Consequently, supports belonging to different cells are orthogonal; no orthogonality is required within a cell. The C_j themselves remain sets of classical instances. They are not subspaces, and the answer sectors $P_j \mathcal{H}_{\mathrm{out}}$ arise only after an oracle and protocol have encoded them. The partition therefore specifies a required zero pattern in the output distinguishability matrix rather than a ready-made physical observable.

We next characterize exact one-query resolution for a family of unitary oracles $\{U_a : a \in \Omega\}$. An arbitrary one-query protocol may prepare a pure state $|\Psi\rangle$ of the query system Q and a reference R, apply $U_a \otimes I_R$, and then use an instance-independent channel and measurement. Write

$$|\Psi_a\rangle = (U_a \otimes I_R) |\Psi\rangle, \qquad \sigma = \mathrm{Tr}_R |\Psi\rangle\langle\Psi|. \tag{3}$$

The output Gram matrix before the final processing is

$$G_{aa'} = \langle \Psi_a | \Psi_{a'} \rangle = \mathrm{Tr}\big(\sigma U_a^\dagger U_{a'}\big). \tag{4}$$

Proposition 1 (block discrimination). The partition Π_g is exactly resolvable with at most one use of $\{U_a\}$, allowing an arbitrary reference, if and only if there is a density operator σ such that

$$\mathrm{Tr}\big(\sigma U_a^\dagger U_{a'}\big) = 0 \quad \text{whenever} \quad g(a) \neq g(a'). \tag{5}$$

Indeed, exact discrimination requires the spans of the output states associated with different cells to be orthogonal, which is equivalent to the indicated pairwise inner products vanishing. Conversely, orthogonal cell spans can be measured by their support projectors. Purification shows that optimizing over σ already includes mixed probes and arbitrary ancillary systems.

Equation (5), with $\sigma \succeq 0$ and $\operatorname{Tr}\sigma = 1$, is the direct block-discrimination form of standard perfect unitary-discrimination conditions [2, 3]; it is not a new general channel-discrimination theorem. Its role here is to connect an answer partition to a specified oracle family. The Gram matrix need only be block diagonal according to Π_g; states within one cell may overlap or coincide. Complete identification is the singleton-cell special case, where G must be diagonal.

For bounded error, orthogonality is replaced by a common POVM with the required success probability. Multi-query state conversion and function evaluation are characterized by adversary methods [4, 5].

Write $Q_\epsilon(\Pi;\mathcal{U})$ and $R_\epsilon(\Pi;\mathcal{U}_{\rm cl})$ for the optimal quantum and classical worst-case numbers of calls needed with worst-case error at most ϵ, under explicitly matched access conventions, and write $Q_E \equiv Q_0$ for exact quantum complexity. Equation (5) characterizes exact resolvability with at most one oracle use, namely $Q_E(\Pi_g;\mathcal{U}) \le 1$. For every nontrivial partition, zero queries cannot resolve the task, so this is equivalent to $Q_E(\Pi_g;\mathcal{U}) = 1$; the trivial one-cell partition instead has $Q_E(\Pi_g;\mathcal{U}) = 0$. The criterion does not by itself establish an advantage. Speedup additionally requires a gap between Q_ϵ and R_ϵ for the same task and oracle interface. Classically, a deterministic query protocol partitions Ω by its possible transcripts and succeeds exactly only when that transcript partition refines Π_g. A quantum protocol need not create such an intermediate classical transcript; it changes the overlaps among the hypothesis states and measures only after the required cells have separated. The relevant resource is this rate of separation, not the number of formal components in the probe superposition.

The criterion also distinguishes a change of representation from a change of access. If

$$V_a = AU_aB \tag{6}$$

for instance-independent unitaries A, B, then $V_a^\dagger V_{a'} = B^\dagger U_a^\dagger U_{a'} B$; as the probe varies, the attainable Gram matrices are unchanged. More generally, families that exactly simulate one another with $O(1)$ oracle calls have quantum query complexities equal up to constant factors; an approximate simulation must also charge its conversion error to ϵ. Preservation of a claimed quantum–classical gap additionally requires corresponding constant-query classical simulations. Exact one-query behavior can nevertheless change when the response representation is not of the form Eq. (6).

For example, Hadamard conjugation of the answer qubit turns Eq. (7) below into the full controlled-phase action $|x,z\rangle \mapsto (-1)^{zb_x}|x,z\rangle$, so the two forms are exactly equivalent. The restricted sign oracle $P_b|x\rangle = (-1)^{b_x}|x\rangle$ is only a subroutine: $P_{b\oplus 1} = -P_b$, and the complementary truth tables differ by a global phase. Supplying a coherent control for P_b restores the missing relative phase, but coherent control is additional black-box access rather than an automatic operation on an unknown unitary [6].

Deutsch's problem illustrates both the block criterion and this dependence. Let $b = (b_0, b_1)$ be the truth table of a Boolean function and use

$$O_b|x,y\rangle = |x, y\oplus b_x\rangle\,. \tag{7}$$

With the answer qubit in $|-\rangle$, phase kickback gives, up to the common answer factor, the output states on the address register:

$$|\psi_{00}\rangle = -|\psi_{11}\rangle = |+\rangle\,,\ |\psi_{01}\rangle = -|\psi_{10}\rangle = |-\rangle\,. \tag{8}$$

Thus the constant and balanced cells occupy orthogonal rays, while the two instances inside either cell are not distinguished. This is precisely a block-diagonal, but not diagonal, Gram matrix and gives the familiar one versus two exact-query separation [7].

The phenomenon is controlled by the response spectrum. Let V be any unitary and consider the controlled-response family indexed by the same four Boolean truth tables,

$$U_b^{(V)} = \sum_{x=0}^{1} |x\rangle\langle x| \otimes V^{b_x}. \tag{9}$$

Theorem 1 (response spectrum). The Deutsch constant–balanced partition is exactly one-query resolvable through $\{U_b^{(V)}\}$, with arbitrary probes and references, if and only if -1 is an eigenvalue of V.

For necessity, write the probe as $|0\rangle|\phi_0\rangle + |1\rangle|\phi_1\rangle$, with $|\phi_x\rangle$ on the response and reference systems. Let $\rho_x = \operatorname{Tr}_R|\phi_x\rangle\langle\phi_x| \succeq 0$ be its unnormalized response operator and put

$$p_x = \operatorname{Tr}\rho_x, \qquad z_x = \operatorname{Tr}(\rho_x V), \qquad p_0 + p_1 = 1. \tag{10}$$

The four cross-cell orthogonality conditions include their complex conjugates and reduce to

$$z_0 = z_0^* = -p_1, \qquad z_1 = z_1^* = -p_0. \tag{11}$$

Since $|z_x| \le p_x$, these equations force $p_0 = p_1 = 1/2$ and $z_0 = z_1 = -1/2$. Equality in the unitary expectation bound is possible only when the support of each nonzero ρ_x lies in the -1 eigenspace of V. Conversely, a -1 eigenvector provides ordinary phase kickback and Eq. (8).

Related qudit versions of Deutsch–Jozsa use higher-dimensional address and response systems [8]. The theorem above instead holds the four Deutsch instances fixed and varies the response unitary. It is also distinct from prior-distribution "useless-query" bounds, which infer quantum lower bounds from classical queries that reveal no information about a property [9].

Corollary 1 (cyclic response). For the elementary cyclic response $X_d\,|y\rangle = |y+1 \bmod d\rangle$, $d \geq 2$, computational-basis queries return the same Boolean point value for every d (by mapping $y \mapsto y + b_x \pmod d$ with $b_x \in \{0,1\}$), but -1 occurs in the spectrum of X_d exactly when d is even. Therefore

$$Q_E^{(d)} = \begin{cases} 1, & d \text{ even}, \\ 2, & d \text{ odd}, \end{cases} \qquad D^{(d)} = 2. \tag{12}$$

In particular, reversible qutrit addition removes Deutsch's exact quantum advantage without changing its classical point values. This is a constant-factor effect, not an asymptotic separation, but it demonstrates that reversibility and blank-register behavior do not determine coherent query power.

Parity provides a broader illustration of the same principle under standard pointwise access. Let $b \in \{0,1\}^N$ and $O_b\,|i,y\rangle = |i, y \oplus b_i\rangle$. Every pure probe, including a reference system, can be written in the answer qubit's X basis as

$$|\Psi\rangle = \sum_{i=1}^{N} |i\rangle \left(|+\rangle\,|\alpha_i\rangle + |-\rangle\,|\beta_i\rangle\right), \qquad p_i = \|\beta_i\|^2. \tag{13}$$

The overlap between the outputs for 0^N and the odd string e_i is

$$\langle \Psi_{0^N} | \Psi_{e_i} \rangle = 1 - 2p_i. \tag{14}$$

Exact even–odd separation would require $p_i = 1/2$ for every i, while normalization gives $\sum_i p_i \leq 1$. Hence one query is possible only for $N \leq 2$, with Deutsch as the limiting case. More generally, the acceptance probability of a T-query algorithm is a polynomial of degree at most $2T$ in the input bits, whereas parity has exact degree N. Thus $T \geq N/2$, and applying the Deutsch construction to disjoint pairs gives the matching algorithm, so

$$Q_E(\mathrm{Par}_N) = \lceil N/2 \rceil, \qquad D(\mathrm{Par}_N) = N \tag{15}$$

for point-value access [10, 11]. A binary partition can therefore remain query intensive. If instead one declares $\widetilde{O}_b\,|y\rangle = |y \oplus \mathrm{Par}(b)\rangle$ to be a single oracle call, parity becomes one-query resolvable classically as well as quantum mechanically. That does not defeat Eq. (15); it replaces point access by an oracle that already aggregates the desired answer.

The preceding examples point to a common picture: the answer partition specifies the required outputs, whereas the query model determines the cost of separating its cells. We test this picture exhaustively for small point-oracle problems. Let the hidden instance be $b \in \{0,1\}^N$, accessed through Eq. (7) with the address range enlarged to N, and consider every binary answer map having 2^{N-1} instances in each cell. Exchanging the answer labels gives the same unlabeled partition, so their number is

TABLE I. Balanced three-bit tasks up to coordinate permutations, input complements, and answer exchange. The orbit sizes sum to 35.

Representative	Orbit	D	Q_E
x_1	3	1	1
$x_1 \oplus x_2$	3	2	1
Par_3	1	3	2
Maj_3	4	3	2
$x_1 \oplus x_2 x_3$	12	3	2
$\mathrm{Sel}(x_1, x_2, x_3)$	12	2	2

$$M_N = \frac{1}{2}\binom{2^N}{2^{N-1}}. \tag{16}$$

Coordinate permutations, independent complements of coordinates, and answer exchange preserve the point-oracle query cost. They reduce the $M_3 = 35$ partitions to six orbits and the $M_4 = 6435$ partitions to 58. Orbit sizes are restored when reporting the counts below.

For each partition we computed the optimal classical deterministic decision-tree depth D and sought matching bounds on the exact quantum query complexity Q_E. For the Boolean response, one-query feasibility reduces to a finite linear system with rational coefficients, which was solved using exact rational arithmetic. Multiquery feasibility was formulated with the Gram-matrix semidefinite program of Barnum, Saks, and Szegedy [12]. Constructive upper bounds were retained as adaptive trees whose nodes read either one bit or the parity of two bits; the latter is one Deutsch query. Such trees are exact upper certificates, but they are not a general characterization of exact quantum algorithms [13].

For $N = 3$, the exact one-query tests and explicit trees meet for every task; the degree-three representatives also have matching polynomial lower bounds. Table I shows every orbit, rather than only aggregate counts. Here $\mathrm{Sel}(c, u, v) = u$ for $c = 0$ and v for $c = 1$.

Thus 20 of the 35 balanced tasks admit an exact advantage: three by a factor of two and seventeen by a factor of 3/2. The certificates also display the conditional structure hidden by the bare partition. For example, three-bit majority is decided by first obtaining $b_1 \oplus b_2$; if it is zero one reads b_1, and if it is one one reads b_3. Likewise, $F = b_1 \oplus b_2 b_3$ is decided by first reading b_2, followed by b_1 when $b_2 = 0$ and by the Deutsch parity $b_1 \oplus b_3$ when $b_2 = 1$. Both real multilinear polynomials have degree three, so the polynomial method excludes one query and proves these two-query constructions optimal.

For $N = 4$, exact one-query lower obstructions and explicit trees settle 395 partitions: 4, 6, 48, 248, and 89 have $(D, Q_E) = (1,1), (2,1), (2,2), (3,2)$, and $(4,2)$, respectively. Of these, 343 have an exact advantage. For the remaining 43 orbit representatives, covering 6040 partitions, exact trees give $Q_E \leq 3$, while two SCS (Splitting Conic Solver) runs at requested tolerances 10^{-7} and 10^{-8} report the two-query SDP (semidefinite programming) infeasible. The resulting assignments, 1536 at $(D, Q_E) = (3, 3)$ and 4504 at $(4, 3)$, are therefore *numerical candidates*, not exact lower bounds. If accepted, they give candidate totals of 4847 speedup and 1588 no-speedup partitions. Exact dual certificates are required before these totals can be stated as theorems.

Numerical SDP studies of all Boolean functions through four bits were already reported in Ref. [13]; subsequent structural work on exact two-query algorithms also uses that four-bit numerical classification [14]. We cross-validated our orbit representatives against the published NPN (Negation-Permutation-Negation) identifiers of Ref. [13]. After removing dummy variables, its one- and two-query entries agree with all 15 representatives certified here. For the 43 remaining representatives, its reported optimal two-query success probabilities lie between 0.900 and 0.986, consistent with our infeasibility results but still numerical. This agreement checks the enumeration and symmetry bookkeeping; it does not manufacture an exact lower bound.

The narrower contribution of the census is therefore to reorganize balanced functions as unlabeled answer partitions, quotient only by symmetries that preserve point access, attach simple exact certificates where available, and keep numerical status explicit. The observed proportions are finite-model facts, not an asymptotic speedup criterion.

Bernstein–Vazirani provides a complementary access-model example. For $a, x \in \mathbb{F}_2^n$, let $f_a(x) = a \cdot x \pmod 2$. With ordinary phase access, the corresponding character states are orthogonal, so the singleton partition is resolved in one query [15]. We now add an inaccessible address record while preserving every computational-basis value returned by the oracle.

Choose $S \subseteq \{1, \ldots, n\}$, $|S| = k$, and let x_S denote the corresponding substring. Consider the unitary dilation

$$W_a^S \, |x, y, e\rangle = |x, y \oplus a \cdot x, e \oplus x_S\rangle \,. \tag{17}$$

Each call receives a fresh witness in $|0^k\rangle$, which is discarded afterward. The accessible oracle is consequently the channel

$$\Phi_a^S(\rho) = \mathrm{Tr}_E\big[W_a^S(\rho \otimes |0^k\rangle\langle 0^k|)(W_a^S)^\dagger\big] \,. \tag{18}$$

Basis queries still return $a \cdot x$ exactly, while the witness records selected address information and removes coherences between sectors with different x_S, as in standard which-way interference [16]. Define the surviving answer partition

$$\Pi_S = \{C_u : u \in \mathbb{F}_2^{n-k}\}, \qquad C_u = \{a : a_{\bar{S}} = u\}. \tag{19}$$

The one-query statement below concerns this surviving partition Π_S, whose cells are indexed by $a_{\bar{S}}$; it is therefore an access-induced change of the task–oracle pair rather than a same-partition comparison for the original singleton task. For the classical comparison, let $\mathcal{O}^{\mathrm{val}} = \{O_a^{\mathrm{val}} : a \in \mathbb{F}_2^n\}$ denote the value-oracle family induced by computational-basis preparation and readout of Φ_a^S: a query chooses x and returns $a \cdot x$. For $k < n$ and worst-case error $\epsilon < 1/2$, these explicitly specified quantum and classical interfaces give

$$Q_\epsilon^\Phi(\Pi_S) = 1, \qquad R_\epsilon(\Pi_S; \mathcal{O}^{\mathrm{val}}) = n - k. \tag{20}$$

For the quantum part, the usual Bernstein–Vazirani input and phase kickback, followed by tracing out the witness, leave

$$\rho_a^S = 2^{-n} \sum_{x, x' : \, x_S = x'_S} (-1)^{a \cdot (x \oplus x')} \, |x\rangle\langle x'| \,. \tag{21}$$

Character orthogonality after the final Hadamards gives, up to a fixed reordering of coordinates,

$$\begin{gathered} H^{\otimes n} \rho_a^S H^{\otimes n} = |a_{\bar{S}}\rangle\langle a_{\bar{S}}| \otimes \frac{I_S}{2^k}, \\ \Pr(z|a) = 2^{-k} \, \mathbf{1}[z_{\bar{S}} = a_{\bar{S}}]. \end{gathered} \tag{22}$$

Thus the one-query readout exactly resolves Π_S. Zero queries cannot resolve this nontrivial partition Π_S, because the output would be independent of its cell label. The classical upper bound uses the basis queries outside S. For the lower bound, impose the favorable promise $a_S = 0$. Under the uniform distribution on $a_{\bar{S}}$, fewer than $n - k$ returned linear equations leave an affine solution space of positive dimension, so even a decoder given the complete transcript succeeds with probability at most $1/2$. The same average bound holds after including the protocol's internal randomness, and therefore some input has success at most $1/2$. If the witness were coherently available, the fixed coupling $|x, e\rangle \mapsto |x, e \oplus x_S\rangle$ could be undone without another oracle call, restoring the ordinary Bernstein–Vazirani output. The accessible oracle here is a channel rather than the globally unitary dilation, highlighting the distinction between standard and garbage-bearing oracle access in quantum query complexity [17].

At $k = 0$, no address coordinate is recorded and Eq. (22) is the ordinary Bernstein–Vazirani output resolving the singleton partition, giving the n-versus-one identification gap. At $k = n$, the stated interference readout is uniform and Π_S has only one cell, although ordinary basis queries can still recover a. The model separates loss of the phase-interference route from loss of the classical function values themselves.

Machine-checked companion. The supplementary archive includes a Lean 4/mathlib formalization of the exact algebraic and finite core. Three `native_decide` audits reproduce the 35-task and 6,435-task deterministic/restricted-tree censuses, including the 395 exact-certificate cases, while additionally trusting Lean's compiler and native evaluator. Density-operator/PVM and unrestricted query semantics, the general parity lower bound, and exact lower bounds for the 6,040 candidates remain unformalized.

We conclude by returning to the role of the answer partition. Equation (1) specifies which hidden instances share an answer, while the oracle and protocol provide the physical ingredients—the probe state, oracle transformation, accessible output space, and measurement—that resolve the corresponding answer classes. Together, these ingredients determine how rapidly states from different cells separate. The response-spectrum theorem and the contrast between point and aggregate parity show how changing the access model can change this separation even when the abstract task remains fixed. Conversely, the finite census holds the point oracle fixed while varying the balanced target. Its exact three-bit certificates show how quantum advantages can arise from oracle-accessible, conditionally composed refinements, while the four-bit calculation clearly separates exact certificates from numerical candidates.

Shor's and Grover's algorithms fit the same task–oracle perspective: Shor exploits periodic structure in the oracle family accessible via the quantum Fourier transform, while Grover exploits symmetry and coherent amplitude amplification via a matched phase oracle. These examples reinforce that quantum speedup depends on the structure of the oracle access, rather than on the coarseness or cardinality of the answer partition alone.

Partitions are therefore indispensable as specifications but insufficient as complexity criteria. Quantum advantage is not caused by partition coarseness or by simultaneous access to imagined branch values. It occurs when structure-matched interference separates the required answer cells in fewer calls than any classical transcript under the same interface. Superlinear exact separations for total Boolean functions are known [18], but establishing any asymptotic speedup requires a structured family of task–oracle pairs; percentages from a finite census cannot substitute for that analysis.

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Data and code availability.—The Python/Lean 4 sources, proof report, certificates, witnesses, tables, and NPN cross-validation are in the supplementary archive.

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